

Assignment #2

Stock Price Forecasting using RNN and LSTM

Submitted By: Muhammad Usman Noor (Reg: 539240)

Course: Generative AI and Applications

Overview

This assignment explores time-series forecasting using Recurrent Neural Networks (RNN) and Long Short-Term Memory Networks (LSTM). The dataset used for prediction consists of Apple's (AAPL) daily closing stock prices from 2015 to 2024. The goal is to use the past 60 days of stock prices to predict the next day's price.

The dataset contains almost ten years of daily historical closing prices for Apple Inc. (AAPL). After cleaning and processing, only the 'Close' column was used. The data was normalized using MinMaxScaler to improve model performance and stability.

Performance Metrics

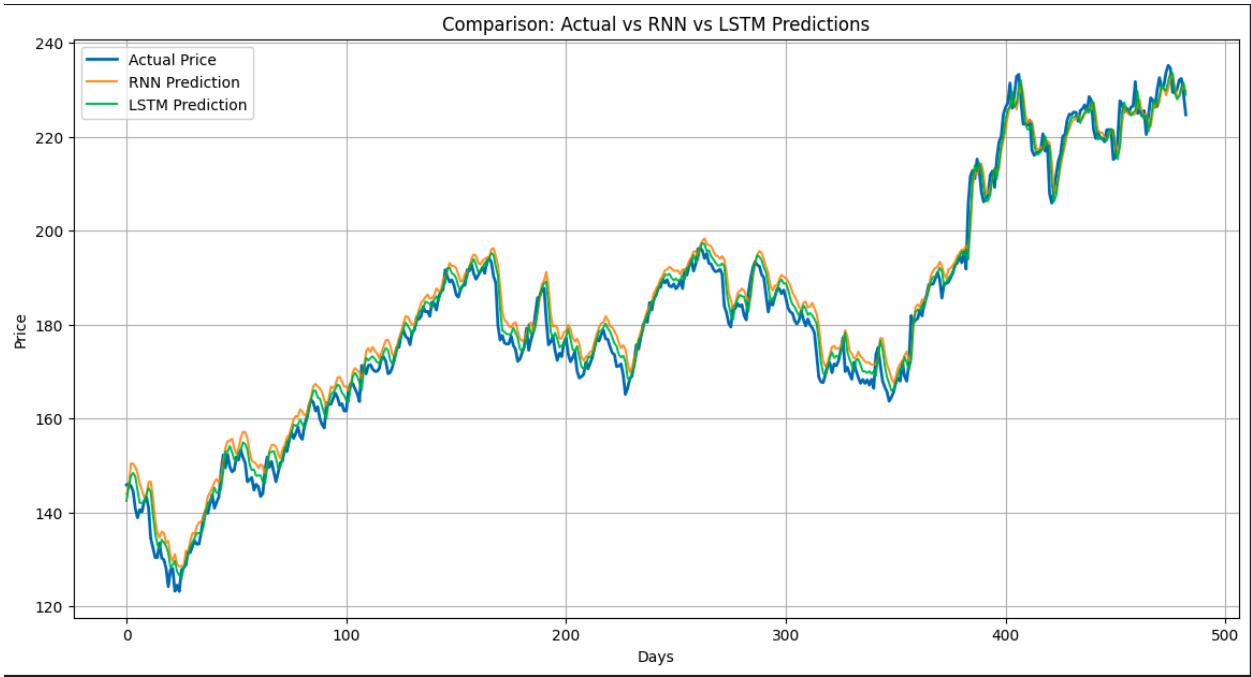
Model	MAE	RMSE
RNN	2.558	3.081
LSTM	1.214	1.499

Model Performance Discussion

The LSTM model achieves significantly lower MAE and RMSE values compared to the Vanilla RNN. This is because LSTMs contain memory cells and gating mechanisms that allow them to retain long-term patterns in sequential data, while RNNs suffer from vanishing gradients and forget earlier information.

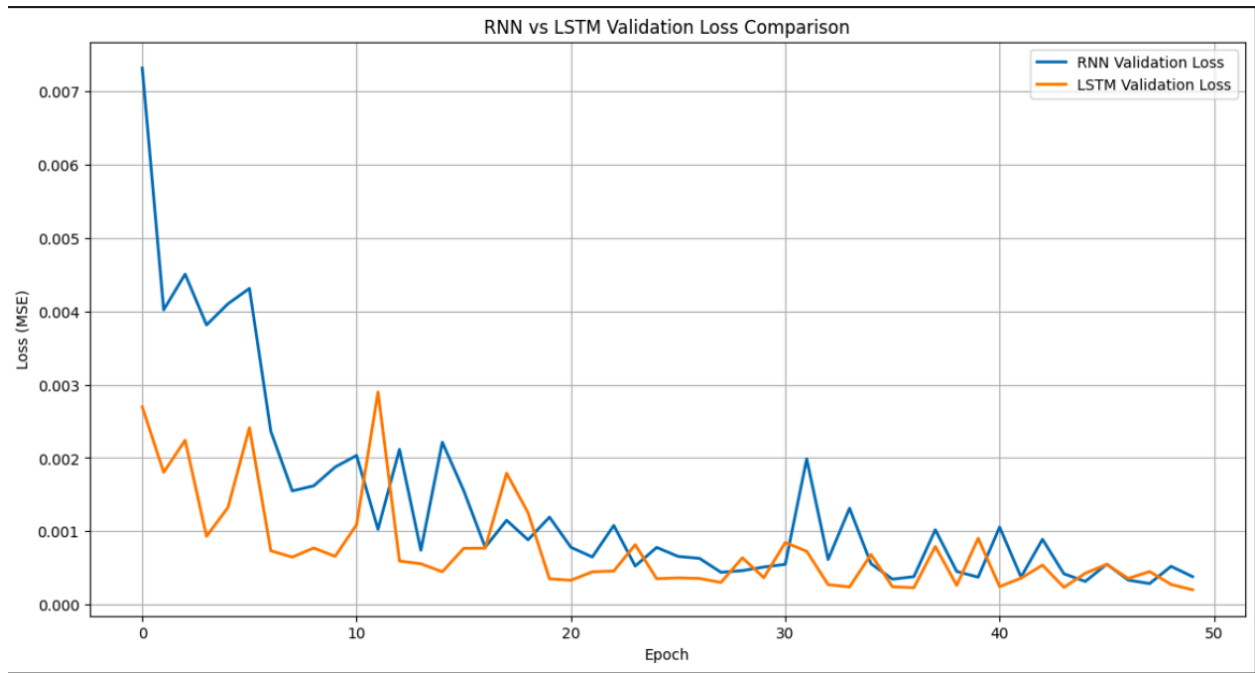
As a result, RNN predictions tend to lag behind actual prices and show more fluctuation. LSTM predictions, on the other hand, track the true market trend more closely and generalize better in unseen data.

Actual vs Prediction:



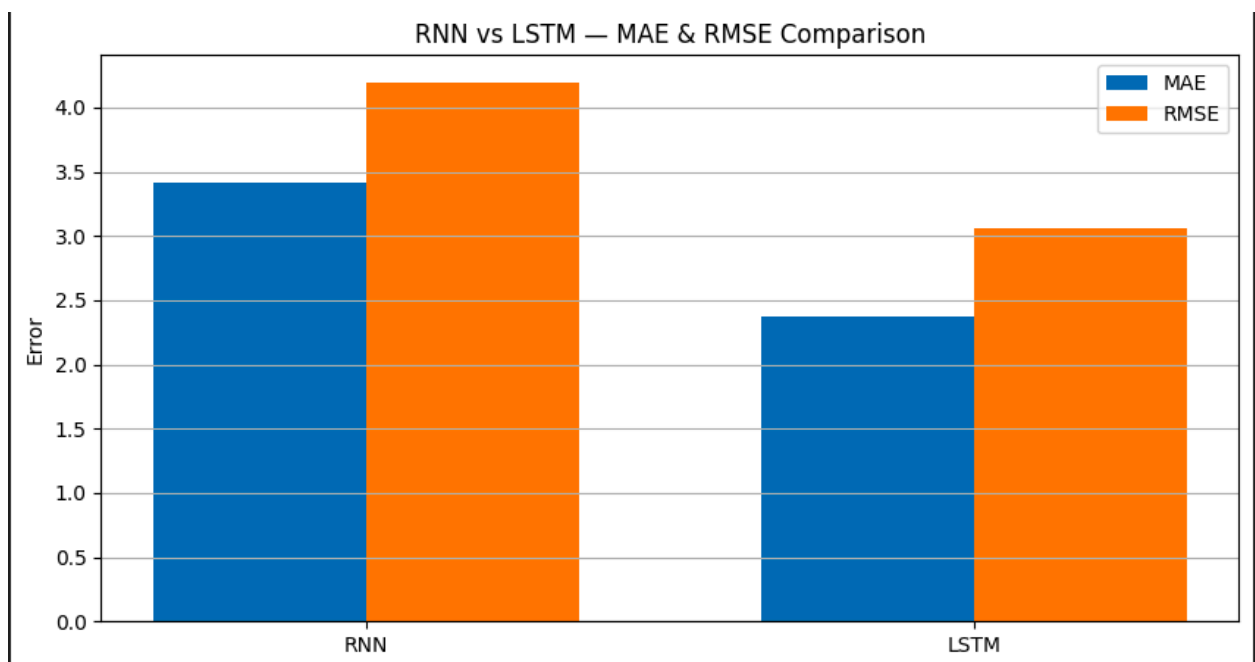
The figure above shows that the LSTM model stays closer to the actual stock price trend compared to the RNN model. The RNN curve shows more deviation due to weaker memory capabilities

Training Loss Curves



The loss curves demonstrate the learning behavior of both models. LSTM achieves a smoother and consistently lower loss, indicating better convergence and generalization. RNN shows a slower decrease and higher overall loss.

RNN vs LSTM Losses Comparison:



Conclusion

This assignment demonstrates that LSTM models outperform traditional RNN models in time-series prediction tasks requiring long-term memory. Their ability to maintain temporal dependencies over many timesteps makes them ideal for stock forecasting, where past market behavior influences future trends. Future improvements may include adding more features (volume, indicators), experimenting with GRUs, or applying attention-based sequence models for enhanced predictive power.

